

Smart Energy Meter - Pin Connection Table

ESP32 pin map used by the current firmware version.

Pin Map

Part / Module	ESP32 Pin	Signal / Use	Notes
ACS712 current sensor OUT	GPIO34	Current analog input	Input only ADC pin
ZMPT101B voltage sensor OUT	GPIO35	Voltage analog input	Input only ADC pin
OLED SDA	GPIO21	I2C data	OLED address 0x3C
OLED SCL	GPIO22	I2C clock	OLED address 0x3C
Relay module IN	GPIO18	Relay control	Code uses active-low relay logic
Relay push button	GPIO19	Manual relay ON/OFF button	INPUT_PULLUP
Wi-Fi setup button	GPIO27	Setup/reset Wi-Fi mode	INPUT_PULLUP
Buzzer + driver	GPIO23	Alert output	Use driver transistor if buzzer needs more current
All low-voltage modules	GND	Common ground	ESP32, sensors, relay module, OLED share low-voltage
Low-voltage supply	5V / 3.3V	Module supply	Use correct voltage required by each module

Notes

- GPIO34 and GPIO35 are input-only ADC pins.
- Relay logic in the code is active-low. Check your relay module before powering a load.
- All low-voltage module grounds must be common.
- AC mains wiring must be enclosed and checked by a qualified person.